The Rosetta Stone: The Reader/Writer Mapping Formulas

This document outlines the two fundamental transformations for re-rendering an existing SVG chart onto a new coordinate system by re-calculating the viewbox coordinates for every point.

1. The Forward & Inverse Mapping Formulas

These two formulas are the building blocks for all transformations. They define the relationship between the **Data Space** and the **Viewbox Space**.

Definitions:

- d_i: A data point (e.g., yield).
- v_i: The corresponding pixel coordinate in the viewbox.
- $[d_{min}, d_{max}]$: The data range of the axis.
- $[vb_{min}, vb_{max}]$: The pixel range of the axis in the viewbox.

Forward Mapping (Data to Viewbox):

> $$v_i = vb_{max} + (d_i - d_{min}) \cdot \frac{vb_{min} - vb_{max}}{d_{max} - d_{min}} \quad (1)$$

Inverse Mapping (Viewbox to Data):

> $$d_i = d_{min} + (v_i - vb_{max}) \cdot \frac{d_{max} - d_{min}}{vb_{min} - vb_{max}} \quad (2)$$

2. The Two Transformation Cases

Case 1: General Translation & Scaling

This transformation re-maps the data to a new data window defined by new minimum and maximum values. This handles both shifting and stretching/squashing the axis.

Transformation Definition:

- The original mapping is defined by $[d_{min}, d_{max}]$.
- The new mapping is defined by $[d'_{min}, d'_{max}]$.

Solution:
To find the new pixel coordinate v'_i for an old pixel coordinate v_i, we must first find the data point d_i using the **Inverse Mapping** on the old system, and then apply the **Forward Mapping** with the new parameters.

1. **Deconstruct:** Find d_i from v_i using Formula (2):

   > $$d_i = d_{min} + (v_i - vb_{max}) \cdot \frac{d_{max} - d_{min}}{vb_{min} - vb_{max}} \quad (3)$$

2. **Reconstruct:** Find v'_i from d_i using Formula (1) with the new parameters:

   > $$v'_i = vb_{max} + (d_i - d'_{min}) \cdot \frac{vb_{min} - vb_{max}}{d'_{max} - d'_{min}} \quad (4)$$

Case 2: Pure Rotation

This transformation performs a geometric rotation of the data window around the pivot point $(x_{d_{min}}, y_{d_{min}})$ by a counter-clockwise angle θ. This is achieved by defining a new set of data mapping parameters.

Transformation Definition:

- **Pivot Point:**

> $$x'_{d_{min}} = x_{d_{min}} \quad (5a)$$
>
> $$y'_{d_{min}} = y_{d_{min}} \quad (5b)$$

- **New Maximums (Derived from rotating the data window):**

> $$x'_{d_{max}} = x_{d_{min}} + (x_{d_{max}} - x_{d_{min}})\cos(\theta) - (y_{d_{max}} - y_{d_{min}})\sin(\theta) \quad (6a)$$
>
> $$y'_{d_{max}} = y_{d_{min}} + (x_{d_{max}} - x_{d_{min}})\sin(\theta) + (y_{d_{max}} - y_{d_{min}})\cos(\theta) \quad (6b)$$

Solution:

The solution process is identical to Case 1. We deconstruct the original points to find the data point (x_{d_i}, y_{d_i}) and then reconstruct the new pixel coordinates (x'_{v_i}, y'_{v_i}) using the new mapping parameters defined in Formulas (5) and (6).

1. **Deconstruct:** Find (x_{d_i}, y_{d_i}) from (x_{v_i}, y_{v_i}) using the original mapping.
2. **Reconstruct:** Find (x'_{v_i}, y'_{v_i}) from (x_{d_i}, y_{d_i}) using the new mapping parameters.